Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years of production **JavaScript/TypeScript and Python engineering** in distributed, event-driven SaaS systems, specializing in **GenAI integration and applied ML for real estate data**. Built AgentProbe — a full-stack multi-agent AI framework (React, FastAPI, LangGraph, multi-provider LLM routing, 5-metric eval engine) deployed on AWS-compatible infrastructure. Built LeaseI — an agentic RAG pipeline for real estate document analysis (93.6% precision, 87.9% F1 on 1,240 leases). Shipped 20+ production microservices at Entrata (RabbitMQ event-driven architecture, AWS S3/EC2/RDS, Docker, PostgreSQL) across 1M+ property records for 100K+ monthly active users. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 -- Aug 2024

Trainee → Associate Engineer → Software Engineer

- Built and owned **20+ distributed microservices and event-driven modules** (Python, JavaScript/TypeScript, PostgreSQL, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users across 1M+ property records — 99.5% uptime over 6+ consecutive months.
- Shipped Entrata's **first production AI pipeline** (Python, PyTorch, OpenAI API) across 1.6M real financial records with output validation and quality gates — designed end-to-end, delivered to production, later productized into a customer-facing feature.
- Achieved **100% PHI compliance** on AWS S3 migration (Libsodium-encrypted pipelines, zero security incidents); improved query performance 30% across 200+ SQL queries; cut execution time 50% via Docker containerization.
- Developed **affordable housing compliance automation** (LIHTC, Section 8/HCV, HUD) across 100K+ unit portfolios — eliminated 80% of manual workload; mentored 5+ engineers through 300+ peer code reviews.

George Mason University

Aug 2024 -- Aug 2025

Research Associate — Data Engineering · Statistical Modeling · Anomaly Detection

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries; applied Isolation Forest to automate anomaly detection across 3 research pipelines.
- Designed Tableau, ggplot2, and Quarto dashboards for faculty leadership — improved research comprehension across two cohorts by 40%.

PROJECTS

AgentProbe: Full-Stack Multi-Agent AI Framework

Open Source

Python · JavaScript/TypeScript · React · LangGraph · LangChain · FastAPI · ChromaDB · OpenAI · Anthropic · Groq · Ollama · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built end-to-end: **React + TypeScript observability dashboard** + **15-endpoint FastAPI backend** orchestrating 3 LangGraph agents (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider GenAI routing (OpenAI, Anthropic, Groq, Ollama) — AWS-deployable architecture with Docker containerization and GitHub Actions CI/CD.
- Implemented a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with automated CI/CD quality gates — demonstrates production GenAI integration discipline; PyTorch and scikit-learn for classical ML baselines.

LeaseI: Agentic RAG Pipeline for Real Estate Document Analysis

Python · LangGraph · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an agentic LangGraph RAG pipeline auditing 1,240 residential lease PDFs for compliance gaps across 10 rules, 5 US states + federal standards — **93.6% precision, 82.9% recall, 87.9% F1**; deterministic rules engine keeps every decision auditable — direct proof of GenAI applied to real estate data at CoStar's scale.

CrashClarity: Large-Scale ML Pipeline (104K Records)

Python · R · Scikit-learn · PyTorch · Random Forest · Logistic Regression · Feature Engineering — Live Dashboard

- Achieved **85%+ accuracy** on 104K+ records — Random Forest + Logistic Regression pipeline with feature engineering across weather, road, and time variables; classifier design and model deployment to GitHub Pages interactive dashboard.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering — Sree Venkateshwara College of Engineering — 2013 -- 2017

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL, PL/SQL, R, Bash, HTML/CSS

Backend & Distributed Systems: Node.js, FastAPI, Flask, REST APIs, gRPC, WebSockets, RabbitMQ (event-driven), Microservices, PostgreSQL, MySQL, SQLite, Schema Design, SQL Optimization, Docker, CI/CD (GitHub Actions)

GenAI & ML: LangGraph, LangChain, Multi-Agent Orchestration, RAG Pipelines, LLM Evaluation, LLM-as-Judge, Prompt Engineering, ChromaDB, Vector Databases, OpenAI API, Anthropic API, Groq, Ollama, PyTorch, Scikit-learn, Hugging Face, BERT, spaCy, Embeddings, Anomaly Detection, Feature Engineering

Cloud & Frontend: AWS (S3, EC2, RDS, Glue), React, TypeScript, Tableau, Power BI, Git, Linux, Agile/Scrum (120+ sprints), Playwright